Chapter 04

Exercise Set

1. Express $\frac{(1-i)^2}{-1-i}$ in the form $a+ib$.

2. Evaluate each of the followings:

   (a) $\text{Im}\left\{\frac{1}{1+i}\right\}$ (b) $\left|\frac{z}{\bar{z}}\right|$ (c) $\text{Re}\left\{\frac{z}{\bar{z}}\right\}$.

3. Convert the following numbers into polar form:

   (a) $z=-1-i$, (b) $3-3\sqrt{3}i$, (c) $z=\frac{(1-i)^2}{1+i}$, and (d) $z=\left(\frac{1+i}{1-i}\right)^{16}$.

4. Convert the following numbers into rectangular form:

   (a) $\sqrt{2}\,e^{i\frac{\pi}{4}}$, and (b) $4\,e^{i\frac{\pi}{3}}$.

5. Find the principle argument of the followings:

   (a) $z=(-1-i)^{12}$, and (b) $z=\left(-2+2\sqrt{3}\,i\right)^3$

6. Find all values of z for the following equations and also locate these values in the complex plane:

   (a) $z^2+4=0$.

   (b) $z^3-1-\sqrt{3}i=0$.

   (c) $z^3=-i$.

   (d) $z^5-1=0$.

7. Describe and graph the locus represented by each of the followings:

   (a) $|z-1+2i|>5$.

   (b) $1<|z-5+i|\leq 2$.

   (c) $\text{Im}\{z^2\}=25$.

   (d) $|z|\leq 3$.

   (e) $\text{Re}\{z^2\}<4$.

   (f) $\frac{\pi}{3}\leq \arg z\leq \frac{\pi}{2}$.